

PS5: Does a Covered Call Fund Beat Holding Its Shares?

CHEME 4/5660 Cornell University | Mathematical Companion

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Training only. This assignment uses historical option quotes, a simple price model, and simplified trading assumptions for teaching. This assignment provides no investment advice.

Covered calls. A covered call fund keeps the full loss of its shares, less the premiums, and gives up every gain above each month's strike. A collar limits each month's loss with a put, at the cost of that put's premium.

Simplifying assumptions. One call contract (and, for the collar, one put contract) per 100 shares held, with the premiums in a cash account that earns, or on a loan pays, the risk-free rate. Every option is held to expiration and settled at the close that day, with no early exercise. We omit commissions, dividends, and taxes. The quotes come from an indicative data feed.

The central question

Does a covered call fund beat holding the shares, and what does it give up?

- Study 1: the prediction. Simulate seven months of share prices for each ticker, price the calls on the binomial tree, and compare the distributions of the growth rates of the fund and of holding the shares and cash with no calls.
- Study 2: what happened. Run the fund on each ticker's seven real monthly calls at their market prices, and compare the fund with holding the shares, SPY, and cash over the same days.
- Study 3: the conclusion. Decide what the fund gave up and what the fund got, and what seven months of history can tell us.

As in the course example, S_j is the share price at the start of month j, K_j the calls' strike, $\mathcal{P}_{c,j}$ their premium (USD/share), N_S the fund's shares, and C_j the cash (USD).

Reading guide Both tracks do Studies 1--3. The collar fund on slide 11 is Advanced only.

Pricing the monthly call

The call sold at the start of month j, strike K_j, expires after Δt years. Pricing uses the Cox–Ross–Rubinstein (CRR) binomial tree with $h = 60$ steps of length $\tau = \Delta t / h$:

$$u = e^{\sigma\sqrt{\tau}}, \qquad d = \frac{1}{u}, \qquad \bar{p} = \frac{e^{g_y \tau} - d}{u - d},$$

where σ is the pricing volatility and g_y the continuously compounded risk-free rate. Rolling the payoff $(S - K_j)^+$ back through the tree under $\bar{p}$, discounting by $e^{-g_y \tau}$ per step and keeping at each node the larger of that value and the exercise value (American), gives $\mathcal{P}_{c,j}$.

In code: `build(MyAmericanCallContractModel, ...)` with strike K_j, then `binomial_premium(call; method = :crr, h = 60, ...)` with time Δt, volatility σ, share price S_j, and rate g_y.

No dividends An American call on a stock without dividends is never worth exercising early, so the price matches a European call.

One month of the covered call fund

The fund holds N_S shares (one call contract per 100) and cash C_j, so $W_j = N_S S_j + C_j$ before the fund sells the calls. The premium goes into the cash, which grows by $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$ over the calls' life T; on exercise the fund sells its shares at K_j and buys new ones at S_{j+1}:

$$C_{j+1} = \underbrace{\mathcal{D}_{T,0}(g_y) \cdot \left(C_j + N_S \mathcal{P}_{c,j}\right)}_{\text{cash and premium grown}} - \underbrace{N_S \left(S_{j+1} - K_j\right)^+}_{\text{new shares less strike}}.$$

With $W_{j+1} = N_S S_{j+1} + C_{j+1}$ and $S_{j+1} - (S_{j+1} - K_j)^+ = \min(S_{j+1}, K_j)$:

$$\frac{W_{j+1}}{W_j} = \frac{N_S \left[\min\left(S_{j+1}, K_j\right) + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{c,j}\right] + \mathcal{D}_{T,0}(g_y) \cdot C_j}{N_S S_j + C_j}.$$

Test month $N_S = 1000$, $S_j = 100$, $C_j = 5000$, $K_j = 102$, $\mathcal{P}_{c,j} = 0.87$, $\mathcal{D} = 1.00345$: 1.0052 at 99.66, 1.0275 at 110.

The growth rate over a roll

Over N months that last T_{roll} years in all, the fund's growth rate is the log of final wealth over initial wealth, divided by the time. Multiplying the wealth ratios gives final wealth divided by initial wealth, so:

$$g = \frac{1}{T_{\text{roll}}} \ln\left(\frac{W_{N+1}}{W_1}\right) = \frac{1}{T_{\text{roll}}} \sum_{j=1}^{N} \ln\left(\frac{W_{j+1}}{W_j}\right) \qquad \text{(units: } 1/\text{yr)}.$$

Holding the shares (the same shares and opening cash, selling no calls) uses the same formula with the wealth ratios of holding. The sum does not depend on the order of the ratios, and a ratio of exactly 1 adds nothing.

- In the model: $N = 7$ months of $\Delta t = D_1/365$ year, the first trade's term (D_1 calendar days), so $T_{\text{roll}} = 7\,\Delta t$.
- In history: T_{roll} is the calendar days from a ticker's first entry to its last expiration, divided by 365, about 0.55 year. A gap between two trades (BMY, PFE) adds one more ratio to the sum.

The model of the share price

For each ticker, the model simulates the share price at the start of each month with geometric Brownian motion (GBM), our belief about the share price:

$$S_{j+1} = S_j \exp\left[(\hat{\mu} - \tfrac{\hat{\sigma}^2}{2})\Delta t + \hat{\sigma}\sqrt{\Delta t}\, Z_j\right], \qquad Z_j \sim \mathcal{N}(0,1),$$

on 10,000 paths from $S_1 = S(0)$, the close on the first entry date, with $\Delta t = D_1/365$, the first trade's term of D_1 calendar days. The drift $\hat{\mu} = \hat{g} + \hat{\sigma}^2/2$ and volatility $\hat{\sigma}$ come from the 252 daily changes before that date ($\hat{g}$: mean log growth rate).

The target strike is $m\,S_j$, with $m = K/S(0)$ from the first trade, and the fund sells the listed strike at or above $m\,S_j$ (next slide). The calls are priced at one of two volatilities:

- historical, $\sigma = \hat{\sigma}$, as in the course example;
- implied, $\sigma = \sigma_{\text{IV}}$, implied by the call's market price on the first entry date.

Two probabilities Paths use our belief (drift $\hat{\mu}$); premiums use the risk-neutral tree (drift g_y).

Listed strikes

Listed options are not FLEX options: their strikes lie on a grid, ΔK apart. For each ticker, ΔK is the ticker's most common spacing near the share price in the course options archive, from 0.50 USD (PFE) to 5 USD. Each month the fund sells the smallest listed call strike at or above the target $m\,S_j$, and the Advanced collar fund buys the largest listed put strike at or below $m_1 S_j$:

$$K_j = \Delta K \left\lceil \frac{m\,S_j}{\Delta K} \right\rceil, \qquad K_{1,j} = \Delta K \left\lfloor \frac{m_1 S_j}{\Delta K} \right\rfloor,$$

where $\lceil \cdot \rceil$ rounds up and $\lfloor \cdot \rfloor$ rounds down to a whole number. The ratio K_j / S_j changes from month to month, so the premium is not a fixed fraction of the share price: the report prices the calls on the binomial tree every month on every simulated path. *Strike choice in the model* repeats the fund with the targets in `moneyness_grid`.

Approximation Exchanges vary the spacing with the share price; one spacing per ticker simplifies the model. Study 2 uses the real strikes.

Running the fund on the real trades

Trade j enters at the close $S_{0,j}$, sells calls with strike K_j at their mid price $\mathcal{P}_{c,j}$, and settles at the close $S_{T,j}$, D_j calendar days later. `covered_call_month` runs the trade with $\mathcal{D}_j = e^{g_y D_j/365}$, and the cash carries to the next trade. When trade $j+1$ enters G_j days after trade j expires, with cash $C_{T,j}$, the fund holds its shares and cash:

$$\frac{W_{0,j+1}}{W_{T,j}} = \frac{N_S S_{0,j+1} + e^{g_y G_j/365}\, C_{T,j}}{N_S S_{T,j} + C_{T,j}}.$$

The risk-free rate comes from the 13-week bill yield $y = 0.0376$ and term $\tau = 91/365$, as in the course example:

$$g_y = \frac{1}{\tau} \ln\left(1 + y\,\tau\right) \approx 0.0374 \text{ per year.}$$

Benchmarks Holding: the same shares and opening cash, no calls. SPY: held from the ticker's first entry to its last expiration. Cash: g_y.

Simulation versus real

Here the simulation follows ticker i's real roll: each trade keeps its real target $K_j/S_{0,j}$, its D_j, and its call's implied volatility, with the gaps. The share prices are simulated; each simulated call takes the listed strike at or above the target, priced on the tree at that implied volatility. The simulation gives $N_p = 10{,}000$ pairs $(g_{\text{fund}}^{(p)}, g_{\text{hold}}^{(p)})$, and with the indicator $\mathbf{1}[\cdot]$:

$$\hat{P}_i = \frac{1}{N_p} \sum_{p=1}^{N_p} \mathbf{1}\Big[g_{\text{fund}}^{(p)} > g_{\text{hold}}^{(p)}\Big], \qquad \text{expected count} = \sum_{i=1}^{29} \hat{P}_i.$$

For one ticker, with the real difference $\delta_i = g_{\text{fund}} - g_{\text{hold}}$, the model percentile is:

$$q_i = \frac{1}{N_p} \sum_{p=1}^{N_p} \mathbf{1}\Big[g_{\text{fund}}^{(p)} - g_{\text{hold}}^{(p)} \le \delta_i\Big].$$

Reading A percentile near 0% or 100% means what happened was rare under the model.

Advanced: one month of the collar fund

The collar fund also buys puts with strike $K_1 < K_2$, the calls' strike. With the put's premium $\mathcal{P}_1$ and the call's $\mathcal{P}_2$, the net premium goes into the cash, the cash grows, and both options settle in cash; on a put exercise the fund sells its shares at K_1 and buys new ones at S_{j+1}:

$$C_{j+1} = \mathcal{D}_{T,0}(g_y) \cdot \left(C_j + N_S\left(\mathcal{P}_2 - \mathcal{P}_1\right)\right) - \underbrace{N_S\left(S_{j+1} - K_2\right)^+}_{\text{calls exercised}} + \underbrace{N_S\left(K_1 - S_{j+1}\right)^+}_{\text{puts exercised}}.$$

The shares less the calls' payoff plus the puts' payoff lie between a floor and a cap:

$$S_{j+1} + \left(K_1 - S_{j+1}\right)^+ - \left(S_{j+1} - K_2\right)^+ = \min\Big(\max\left(S_{j+1}, K_1\right),\, K_2\Big),$$

and $W_{j+1}/W_j = (N_S S_{j+1} + C_{j+1})/(N_S S_j + C_j)$. The put is the trade's `long_put` leg, near 0.10 delta.

Course example *A Protective Collar on SPY* uses the same floor K_1, cap K_2, and premiums $\mathcal{P}_1$ (put) and $\mathcal{P}_2$ (call).

Summary

PS5 runs a covered call fund twice: in a model of the share price, and on the real monthly calls of 29 tickers, and compares the fund with holding the shares.

Key takeaways

- Each month the premium goes into the fund's cash, the cash grows, and an exercise is paid from the cash; the wealth ratio is the shares plus cash at the end over the start. The premiums offset share losses, and the calls give up gains above the strike.
- The growth rate over a roll is the sum of the logs of the wealth ratios divided by the time in years, in 1/yr for the model and for history.
- The model's probability that the fund beats holding covers many possible seven-month periods. History gives one period, shared by all 29 tickers.

Use these measures to answer the central question.